

Repository Structure, Application Architecture, and Data Models

1. Repository Structure

treasury-label-verifier/

|

|— README.md

|— .gitignore

|— LICENSE

|— requirements.txt

|— app.py

|

|— docs/

| |— project-charter.md

| |— project-design-document.md

| |— technical-planning-document.md

| |— architecture.md

| |— assumptions-and-limitations.md

| |— test-plan.md

|

|— sample-labels/

| |— valid/

| |— invalid/

```
| | — poor-quality/
| | — README.md
|
| — sample-data/
| | — valid-batch.csv
| | — mixed-errors-batch.csv
| | — missing-warning-batch.csv
| | — record-matching-demo.csv
|
| — sample-batches/
| | — clean-batch/
| | — mixed-error-batch/
| | — poor-quality-batch/
|
| — ocr/
| | — ocr_service.py
| | — image_preprocessing.py
|
| — matching/
| | — record_matcher.py
| | — scoring.py
|
| — validation/
| | — validator.py
```

```
| | — field_extractor.py
| | — warning_rules.py
|
| — utils/
| | — text_normalizer.py
| | — csv_loader.py
| | — timer.py
| | — export_results.py
|
| — deployment/
|   | — render.yaml
|   | — deployment-notes.md
```

2. Application Workflow Specification

Primary Workflow

The prototype is built around a batch-first workflow.

Upload Application CSV

↓

Upload Label Images

↓

OCR Extraction

↓

Text Normalization



Record Matching



Field Validation



Recommendation Engine



Results Dashboard



CSV Export

Secondary Workflow

Single Label Test Mode

Purpose:

- Demonstrations
- Testing
- Troubleshooting

Workflow:

Upload Label



OCR



Validation



Results

This workflow is secondary to the batch workflow.

3. CSV Schema Specification

Required Fields

record_id

brand_name

class_type

alcohol_content

net_contents

government_warning

These fields are required for processing.

If any required fields are missing, processing stops and a validation error is displayed.

Optional Fields

bottler_producer

country_of_origin

beverage_type

These fields are included for future extensibility.

The prototype may display or validate them when present but they are not part of the primary compliance workflow.

4. Data Model Specification

ApplicationRecord

Represents a single application record loaded from CSV.

record_id: string

brand_name: string

class_type: string

alcohol_content: string

net_contents: string

government_warning: string

bottler_producer: string | optional

country_of_origin: string | optional

beverage_type: string | optional

OCRResult

Represents raw OCR output.

image_name: string

ocr_text: string

processing_time_seconds: float

RecordMatchResult

Represents the outcome of matching a label to an application record.

record_id: string

matched_record: string

confidence: float

status:

Auto Match

Needs Review

No Match

VerificationResult

Represents a single field validation result.

field: string

expected: string

detected: string

status:

Pass

Fail

Missing

Needs Review

match_type:

Exact

Fuzzy

Normalized Exact

Strict

confidence: float

notes: string

LabelVerificationResult

Represents validation results for one image.

image_name: string

matched_record_id: string

match_confidence: float

overall_status:

Pass

Fail

Needs Review

Error

processing_time_seconds: float

ocr_text: string

results:

VerificationResult[]

BatchVerificationResult

Represents an entire batch.

batch_status:

Completed

Completed With Review

Error

total_files: integer

processing_time_seconds: float

items:

LabelVerificationResult[]

5. Record Matching Rules

The prototype includes a conservative matching strategy.

Auto Match

Confidence:

95% and above

Action:

Automatically associate label with application record.

Needs Review

Confidence:

70%–94%

Action:

Suggest likely record.

Human confirmation recommended.

No Match

Confidence:

Below 70%

Action:

No automatic association.

Manual review required.

6. Validation Status Rules

Pass

The detected field matches the expected value.

Fail

The detected field clearly conflicts with the expected value.

Missing

The required field could not be found.

Needs Review

The application cannot confidently determine the result.

Examples:

- OCR uncertainty
- Ambiguous matching
- Poor image quality
- Missing information

The prototype intentionally favors human review when uncertainty exists.

Error

The image could not be processed.

Examples:

- Unsupported file
 - Corrupt image
 - Processing failure
-

7. Matching Logic

Exact Matching

Used when values must be legally or numerically precise.

Fields:

- Alcohol Content
- Net Contents

Example:

Expected:

45% Alc./Vol. (90 Proof)

Detected:

40% Alc./Vol. (80 Proof)

Result:

Fail

Fuzzy Matching

Used when formatting differences should not cause failure.

Fields:

- Brand Name
- Class / Type

Example:

Expected:

STONE'S THROW

Detected:

Stone's Throw

Result:

Pass

Strict Matching

Used when federal regulations require exact language.

Fields:

- Government Warning

Result:

Fail if required warning language is missing or materially altered.

Normalized Matching

Used when formatting differs but meaning remains identical.

Example:

Expected:

750 mL

Detected:

750ml

Result:

Pass

Normalization may remove:

- Extra spaces
- Case differences
- Limited punctuation differences

Normalization does not ignore actual value differences.

8. CSV Export Format

The prototype supports exporting results as CSV.

Potential export fields:

image_name

record_id

brand_name

overall_status

match_confidence

processing_time_seconds

field

expected_value

detected_value

field_status

notes

Purpose:

- Supervisor review
 - Compliance workflow support
 - Documentation
 - Audit preparation
-

9. Future Production Models

Future versions may introduce additional models for:

- User accounts
- Authentication
- Audit logging
- Workflow routing
- COLAs Online integration
- Document retention
- Large-scale batch processing

These capabilities are intentionally excluded from the prototype scope.